

Hematology Thrombosis / Anticoagulants / TMA Codex Guide

- Block: Hematology
- Method: Codex-authored manual distillation from extracted lecture markdown in this workspace.
- Sources:
 - 20210308_Yui_Overview_Thrombotic_Disorders.pdf
 - 20230306_Yui_Thrombophilic_Syndromes.pdf
 - 20230308_Yui_Anticoagulants.pdf
 - Yui_Thrombotic_Microangiopathies.pdf

One-Minute Mental Model

Thrombosis questions usually ask which side of hemostasis has tipped too far toward clotting. First localize the clot:

- **Arterial thrombi** form in high-pressure vessels, often over ruptured atherosclerotic plaque, and are platelet-predominant “white” thrombi.
- **Venous thrombi** form in low-pressure veins, usually deep leg veins, and are fibrin/RBC-rich “red” thrombi driven by Virchow’s triad.
- **Microvascular thrombi** cause thrombotic microangiopathy: platelets are consumed in tiny vessels, RBCs shear into schistocytes, and organs can fail.

Then decide whether the issue is inherited/acquired hypercoagulability, medication effect, or a disease syndrome such as TTP, HUS, or DIC.

High-Yield Lecture Figures

Virchow’s Triad

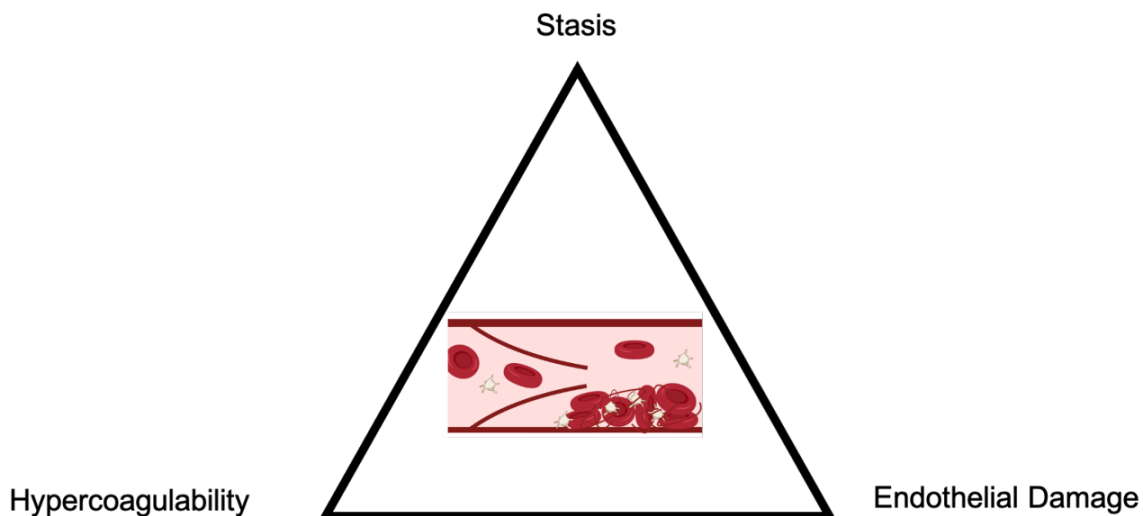


Figure 1: Virchow’s triad: stasis, hypercoagulability, and endothelial damage.

Highest-Yield Concepts

Arterial vs Venous Thrombosis

- Arteries have narrow lumens, thick walls, high-pressure shear flow, and no valves; venous vessels have wider lumens, thin walls, lower pressure, and one-way valves. [Overview Thrombotic Disorders, slide 3]

Anticoagulants

- Heparins
 - Unfractionated heparin
 - Low molecular weight heparin
- Warfarin
- F10a inhibitors
- Direct thrombin inhibitors



Figure 2: Major anticoagulant categories used to interrupt clot formation.

Laboratory Values in TTP, HUS, and DIC

	TTP	HUS	DIC
Hemoglobin	↓	↓	↓
Platelet count	↓↓	↓↓	↓↓
Indirect bilirubin	↑	↑	↑
Haptoglobin	↓	↓	↓
Lactate dehydrogenase	↑↑	↑↑	↑
Creatinine	Normal/↑	↑↑	Normal/↑
PT/PTT	Normal	Normal	↑
Fibrinogen	Normal	Normal	↓
D-dimer	Normal/↑	Normal/↑	↑
ADAMTS13	<10%	Normal	Normal
Schistocytes	Marked	Marked	Mild

Figure 3: Laboratory patterns in TTP, HUS, and DIC.

- Arterial thrombi are platelet-predominant “white” thrombi and commonly occur in cerebral vessels, coronary vessels, and the aorta. [Overview Thrombotic Disorders, slide 3]
- Arterial thrombosis often follows atheromatous plaque rupture, exposing damaged endothelium/tissue factor and causing platelet adhesion. [Overview Thrombotic Disorders, slide 4]
- Venous thrombi are RBC/fibrin-rich “red” thrombi, most often in deep leg veins, pulmonary arteries, superficial veins, or cerebral venous sinus. [Overview Thrombotic Disorders, slides 3 and 5]
- Venous thrombosis is driven by Virchow’s triad: stasis, endothelial damage, and hypercoagulability. [Overview Thrombotic Disorders, slides 5-7]
- Provoked clots have an identifiable trigger. Major transient risks include surgery and immobility; moderate/transient risks include estrogen/OCP exposure and malignancy; age and obesity are non-transient risk factors. [Overview Thrombotic Disorders, slide 8]
- DVT symptoms come from venous obstruction and vascular inflammation: asymmetric swelling and dilated superficial veins from obstruction; pain, erythema, and warmth from inflammation. [Overview Thrombotic Disorders, slides 9-10]
- PE occurs when a venous thrombus embolizes through the IVC/right heart into the pulmonary arteries. [Overview Thrombotic Disorders, slide 12]

Inherited and Acquired Thrombophilias

- Antithrombin normally inhibits thrombin IIa plus factors IXa, Xa, and XIa. Antithrombin deficiency increases thrombin/fibrin generation. [Thrombophilic Syndromes, slide 6]
- Antithrombin deficiency may be congenital or acquired from liver disease or nephrotic syndrome; risk of thrombosis rises about 10-15x; test with antithrombin activity. [Thrombophilic Syndromes, slide 7]
- Protein C and protein S normally inactivate VIIIa and Va, limiting tenase and prothrombinase complexes. Deficiency increases Xa and thrombin generation. [Thrombophilic Syndromes, slide 8]
- Protein C/S deficiency may be congenital or acquired from vitamin K deficiency or liver disease; protein S also falls with pregnancy and estrogen exposure. Test with protein C/S activity. [Thrombophilic Syndromes, slides 9-10]
- Factor V Leiden is a point mutation in F5 that makes factor Va resistant to activated protein C. It is autosomal dominant; heterozygotes have about 5x thrombosis risk and homozygotes about 50x. [Thrombophilic Syndromes, slides 11-13]
- Factor V Leiden testing can be genetic testing for the F5 point mutation or functional testing for activated protein C resistance. [Thrombophilic Syndromes, slide 13]
- Prothrombin G20210A is an autosomal dominant gain-of-function F2 mutation causing increased prothrombin levels and about 3-4x thrombotic risk. [Thrombophilic Syndromes, slide 14]

Antiphospholipid Syndrome

- APS is an autoimmune thrombophilia requiring clinical criteria plus laboratory criteria; it may be associated with SLE and is often called lupus anticoagulant. [Thrombophilic Syndromes, slide 15]
- APS presents with venous thrombosis, arterial thrombosis, and obstetric complications such as early miscarriages or premature birth from pre-eclampsia/placental insufficiency. [Thrombophilic Syndromes, slide 16]
- APS is paradoxical: antiphospholipid antibodies prolong phospholipid-dependent coagulation tests in vitro, but clinically the patient is hypercoagulable in vivo. [Thrombophilic Syndromes, slide 17]
- APS labs can show prolonged aPTT or PT that does not correct on mixing, but PT is often normal because PT assays include more phospholipid. [Thrombophilic Syndromes, slide 20]
- DRVVT uses Russell viper venom to activate factor X in a phospholipid-dependent assay. APS prolongs DRVVT, it does not correct with mixing, but it corrects with excess phospholipid. [Thrombophilic Syndromes, slides 18-20]
- APS testing also includes anticardiolipin and beta-2 glycoprotein IgG/IgM antibodies. [Thrombophilic Syndromes, slide 20]

Anticoagulants

- Anticoagulants inhibit coagulation/fibrin clot formation and are used for thromboprophylaxis and treatment of thrombosis. [Anticoagulants, slide 4]
- UFH binds antithrombin and causes a conformational change that enables inactivation of thrombin IIa and factor Xa. It is given IV or subcutaneous, monitored with aPTT, reversed with protamine, and risks bleeding and HIT. [Anticoagulants, slides 7-9]

- LMWH is subcutaneous, has longer half-life and more predictable kinetics, usually does not need aPTT monitoring, and causes antithrombin-dependent inactivation of Xa. Protamine is the reversal agent. [Anticoagulants, slide 10]
- Warfarin inhibits epoxide reductase, depleting active vitamin K and preventing gamma-carboxylation of II, VII, IX, X, protein C, and protein S. [Anticoagulants, slide 11]
- Warfarin initially creates a hypercoagulable state because protein C and S have shorter half-lives than II, VII, IX, and X; bridge with heparin for at least 5 days when starting warfarin. [Anticoagulants, slide 14]
- Failure to bridge warfarin can cause warfarin-induced skin necrosis from acquired protein C deficiency in skin/subcutaneous tissue, especially in patients with underlying thrombophilia. [Anticoagulants, slide 15]
- Warfarin prolongs PT and aPTT, but PT is more sensitive; INR standardizes PT and typical INR goal is 2.0-3.0. [Anticoagulants, slides 16-17]
- Warfarin risks include bleeding, skin necrosis, and teratogenicity. Reversal is vitamin K for new factor activation and FFP for active factor replacement. [Anticoagulants, slide 18]
- Factor Xa inhibitors are DOACs such as apixaban, betrixaban, edoxaban, and rivaroxaban. They inhibit Xa, require no routine monitoring, and have fewer medication interactions. [Anticoagulants, slide 19]
- Direct thrombin inhibitors inhibit IIa, are not antithrombin-dependent, and can be oral or IV; oral versions do not require monitoring. [Anticoagulants, slide 20]
- tPA converts plasminogen to plasmin and can be administered for rapid thrombolysis in severe thrombosis such as massive PE, acute ischemic stroke with deficits, or acute MI. Major risk is bleeding. [Anticoagulants, slides 21-22]
- Tranexamic acid and aminocaproic acid are antifibrinolytics used when hyperfibrinolysis causes bleeding; they inhibit plasminogen activity and decrease conversion to plasmin. [Anticoagulants, slide 23]

Thrombotic Microangiopathy

- TMA is intravascular hemolytic anemia caused by thrombotic lesions in the microvasculature. RBCs shear past microthrombi, producing microangiopathic hemolytic anemia. [TMA, slides 6-7]
- TMA labs show consumptive thrombocytopenia, intravascular hemolysis, low haptoglobin, high indirect bilirubin, high LDH, plasma/urine hemoglobin, and schistocytes. [TMA, slide 8]
- TTP is a primary TMA caused by ADAMTS13 deficiency. Without ADAMTS13, ultra-large vWF multimers persist, platelets aggregate, and platelet-rich microthrombi cause hemolysis. [TMA, slides 9-11]
- TTP manifestations include severe thrombocytopenia/bruising plus end-organ microthrombi: fever, CNS abnormalities, and mild renal injury. [TMA, slide 12]
- Acquired TTP is usually due to autoantibodies against ADAMTS13. Treatment is plasma exchange, which removes antibody and replaces ADAMTS13, plus immunosuppression with steroids and often rituximab. [TMA, slide 13]
- Caplacizumab binds vWF and prevents platelet GPIb-IX-V binding to vWF, improving platelets and hemolysis; side effect is acquired vWD/bleeding risk. [TMA, slide 14]
- HUS is a TMA caused by Shiga toxin-producing E. coli O157:H7, often from raw/undercooked beef; it causes endothelial dysfunction especially in intestine and kidney. [TMA, slide 15]
- HUS presents most often in young children after bloody diarrhea with thrombocytopenia, MAHA, and predominant renal injury. Management is supportive, often including dialysis. [TMA, slide 17]
- DIC is a TMA plus systemic tissue-factor-driven coagulation activation, always driven by an underlying process such as infection, surgery, trauma, obstetric complications, malignancy, or liver disease. [TMA, slides 18-19]
- DIC forms widespread fibrin thrombi, generating D-dimers/FDPs as clots break down, and consumes platelets and clotting factors, causing both thrombosis and bleeding. [TMA, slides 20-22]
- DIC management is treatment of the underlying condition. [TMA, slide 22]

Must-Know Comparisons

Topic	Key mechanism	Core clues	Test/treatment anchor
Arterial thrombosis	Platelet-rich clot over damaged plaque	Stroke/MI/aorta; high-pressure system	Antiplatelet logic often central
Venous thrombosis	Stasis/endothelial injury/hypercoagulability	DVT/PE; red fibrin/RBC thrombus	Anticoagulation logic central
Antithrombin deficiency	Less inhibition of IIa/IXa/Xa/XIa	VTE tendency; congenital or liver/nephrotic acquired	Antithrombin activity

Topic	Key mechanism	Core clues	Test/treatment anchor
Protein C/S deficiency	Less inactivation of Va/VIIIa	VTE tendency; warfarin skin necrosis risk	Protein C/S activity
Factor V Leiden	Va resists activated protein C	Common inherited thrombophilia	F5 genetic test or APC resistance
Prothrombin G20210A	Increased prothrombin level	Mild-moderate VTE risk	F2 genetic test
APS	Autoimmune thrombophilia	VTE/arterial thrombosis/pregnancy loss	DRVVT, anticardiolipin, beta-2 glycoprotein
TTP	ADAMTS13 deficiency	MAHA, severe thrombocytopenia, CNS, normal PT/aPTT	Plasma exchange, steroids/rituximab, caplacizumab
HUS	Shiga toxin endothelial injury	Bloody diarrhea, child, renal injury	Supportive care/dialysis
DIC	Systemic tissue factor activation	Bleeding + thrombosis, low fibrinogen, prolonged PT/aPTT, high D-dimer	Treat trigger

Drug/class	Target	Monitoring	Reversal / key issue
UFH	Antithrombin-dependent IIa and Xa inhibition	aPTT	Protamine; HIT risk
LMWH	Antithrombin-dependent Xa inhibition	Usually none	Protamine; lower HIT risk than UFH
Warfarin	Vitamin K epoxide reductase	INR	Vitamin K/FFP; bridge initially
Xa inhibitors	Direct Xa inhibition	None routine	DOAC convenience
Direct thrombin inhibitors	Direct IIa inhibition	Oral: none routine	Not antithrombin-dependent
tPA	Plasminogen -> plasmin	Clinical	Bleeding risk
TXA/aminocaproic acid	Antifibrinolysis	Clinical	Preserve fibrin clot

Common Confusions To Avoid

- **APS prolongs clotting tests but causes clots.** The in vitro test artifact does not mean the patient bleeds.
- **HIT causes thrombosis, not just low platelets.** Stop all heparin and give a non-heparin anticoagulant.
- **Warfarin needs a bridge because protein C/S fall early.** Early warfarin alone can be prothrombotic.
- **UFH and LMWH both use antithrombin, but UFH hits IIa and Xa while LMWH is mainly Xa.**
- **TTP and HUS have normal PT/aPTT.** DIC prolongs PT/aPTT because it consumes coagulation factors.
- **TTP is not just “the pentad.”** In practice, severe thrombocytopenia + MAHA + low ADAMTS13 is the key pattern.
- **D-dimer is expected in DIC but is not specific.** Interpret it with fibrinogen, PT/aPTT, platelets, and clinical trigger.
- **Arterial vs venous treatment logic differs.** Arterial clotting is platelet-heavy; venous clotting is fibrin/coagulation-heavy.

Rapid Review

- Virchow triad: stasis, endothelial injury, hypercoagulability.
- Arterial clot: platelet-rich, plaque rupture, high-pressure arteries.
- Venous clot: fibrin/RBC-rich, deep leg veins, DVT/PE pathway.
- Antithrombin deficiency: less inhibition of IIa/IXa/Xa/XIa.
- Protein C/S deficiency: less inhibition of Va/VIIIa.
- Factor V Leiden: factor V resists protein C.
- Prothrombin G20210A: increased prothrombin.
- APS: thrombosis/pregnancy morbidity plus antiphospholipid labs; DRVVT corrects with excess phospholipid.
- UFH: antithrombin-dependent IIa/Xa inhibition, aPTT, protamine.
- Warfarin: vitamin K blockade, INR, bridge, teratogenic, reverse with vitamin K/FFP.
- TTP: ADAMTS13 deficiency, ultra-large vWF, severe thrombocytopenia/MAHA, plasma exchange.
- HUS: Shiga toxin, bloody diarrhea, renal injury, supportive care.
- DIC: tissue factor activation, low platelets/factors/fibrinogen, high D-dimer, prolonged PT/aPTT, treat underlying cause.

Active Recall Questions

1. Compare arterial and venous thrombi by vessel type, composition, and mechanism.
2. What are the three elements of Virchow's triad?
3. Why does a leg DVT cause asymmetric swelling and pain?
4. How do antithrombin, protein C, and protein S normally limit clotting?
5. Why does Factor V Leiden cause activated protein C resistance?
6. What makes APS paradoxical on coagulation tests?
7. How does DRVVT confirm phospholipid-dependent inhibition?
8. Compare UFH and LMWH by target, route, monitoring, and reversal.
9. Why does warfarin create an initial hypercoagulable state?
10. What is warfarin-induced skin necrosis?
11. When would tPA be used, and what is the major danger?
12. How do tranexamic acid and aminocaproic acid reduce bleeding?
13. Define microangiopathic hemolytic anemia.
14. Why does TTP produce platelet-rich microthrombi?
15. How does plasma exchange treat acquired TTP?
16. Why can caplacizumab cause an acquired vWD-like bleeding problem?
17. What makes HUS predominantly renal?
18. Why does DIC cause both clotting and bleeding?
19. How do PT/aPTT, fibrinogen, D-dimer, and ADAMTS13 distinguish TTP, HUS, and DIC?
20. What underlying triggers should you search for when DIC is suspected?